Student Performance Management System

BASIC TABLES-

```
SELECT * FROM Students;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	name	age	course	city
1	Nidhi	21	BCA	Dehradun
2	Rahul	22	BCA	Delhi
3	Aman	20	BCA	Mumbai
4	Priya	21	BCA	Noida
5	Karan	23	BCA	Pune

Students 41

Output

```
SELECT * FROM Courses;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_id	course_name	duration
101	BCA	3 Years
102	BBA	3 Years
103	MCA	2 Years
104	MBA	2 Years
105	BTech	4 Years

Courses 42

Output

Action Output

```
SELECT * FROM Courses;

SELECT * FROM Marks;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	course_id	marks
1	101	88
2	102	76
3	103	69
4	104	91
5	105	72

Marks 43

Output

Action Output

#	Time	Action	M
103	16:43:53	SELECT * FROM Students LIMIT 0, 1000	50
104	16:44:35	SELECT * FROM Courses LIMIT 0, 1000	10
105	16:45:11	SELECT * FROM Marks LIMIT 0, 1000	50

```
SELECT * FROM Attendance;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	attendance_percentage
1	85
2	60
3	78
4	90
5	72

Attendance 44

Output

Action Output

#	Time	Action

QUERIES-

Top Students (ORDER BY)

```
SELECT s.name, c.course_name, m.marks
FROM Students s
JOIN Marks m ON s.student_id = m.student_id
JOIN Courses c ON m.course_id = c.course_id
ORDER BY m.marks DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

name	course_name	marks
Rashmi	MA	95
Anjali	MCom	95
Komal	MTech	94
Kriti	MBA	93
Shreya	MBA	93

Result 45

Output

Action Output

#	Time	Action	Message
105	16:45:11	SELECT * FROM Marks LIMIT 0, 1000	50 row(s) returned
106	16:45:54	SELECT * FROM Attendance LIMIT 0, 1000	50 row(s) returned
107	16:46:26	SELECT s.name, c.course_name, m.marks FROM Students s JOIN Marks m ON s.s...	5 row(s) returned

Low Attendance (WHERE)

```
SELECT s.name, a.attendance_percentage
FROM Students s
JOIN Attendance a ON s.student_id = a.student_id
WHERE a.attendance_percentage < 75;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	attendance_percentage
Rahul	60
Karan	72
Rohit	65
Vikas	70
Arjun	68

Result 46

Output

Action Output

GROUP BY + AVG

```
SELECT c.course_name, AVG(m.marks)
FROM Marks m
JOIN Courses c ON m.course_id = c.course_id
GROUP BY c.course_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_name	AVG(m.marks)
BCA	75.2000
BBA	87.6000
MCA	68.6000
MBA	90.2000
BTech	72.4000

Result 47

Output

Action Output

HAVING

```
SELECT c.course_name, AVG(m.marks) AS avg_marks
FROM Marks m
JOIN Courses c ON m.course_id = c.course_id
GROUP BY c.course_name
HAVING AVG(m.marks) > 75;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_name	avg_marks
BCA	75.2000
BBA	87.6000
MBA	90.2000
MTech	86.4000
MCom	90.8000

Result 48

Output

LEFT JOIN

```
SELECT s.name, a.attendance_percentage
FROM Students s
LEFT JOIN Attendance a ON s.student_id = a.student_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_name	avg_marks
BCA	75.2000
BBA	87.6000
MBA	90.2000
MTech	86.4000
MCom	90.8000

Result 48

Output

COUNT

```
SELECT course, COUNT(*) AS total_students
FROM Students
GROUP BY course;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course	total_students
BCA	5
BBA	5
B.Tech	5
MBA	5
B.Com	5

Result 49

Output

Action Output

SUM

```
SELECT SUM(marks) AS total_marks
FROM Marks;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

total_marks
3985

VIEW

```
SELECT * FROM student_performance;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	course_name	marks	attendance_percentage
Nidhi	BCA	88	85
Arjun	BCA	74	68
Ajay	BCA	65	64
Tarun	BCA	74	74
Lokesh	BCA	75	75

student_performance 51

Output

Action Output

PYTHON-

```python
import mysql.connector
import pandas as pd
import matplotlib.pyplot as plt
```

```python
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="NidhiSilmana@123",
    database="student_db"
)

print("Connected Successfully")
cursor = conn.cursor()
print("Connected to Database")
```

Data Output

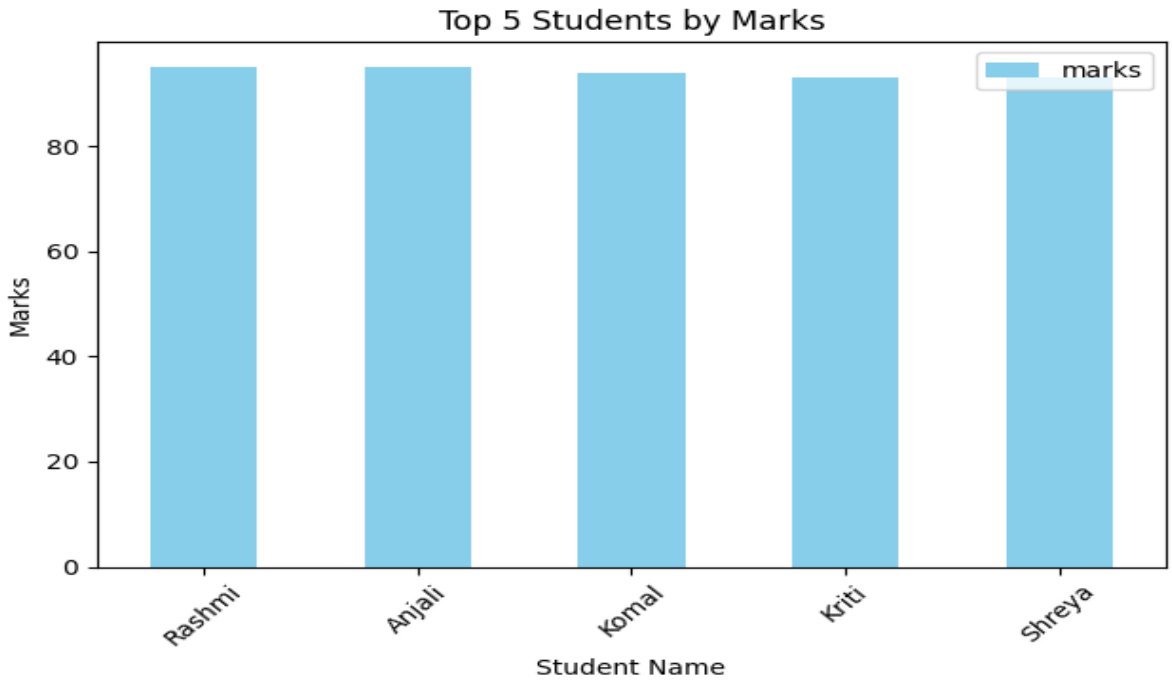
```python
query = "SELECT * FROM Students"
df_students = pd.read_sql(query, conn)
df_students
```

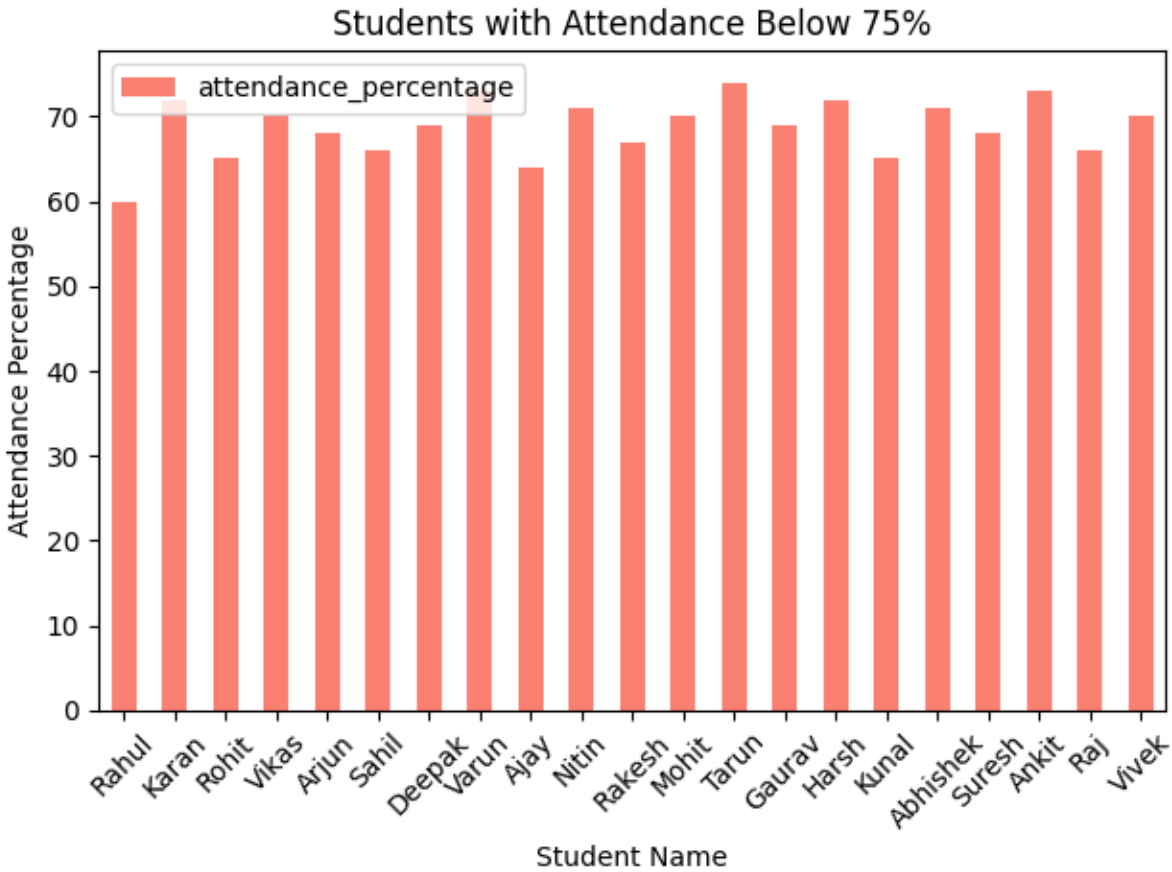
0.0s

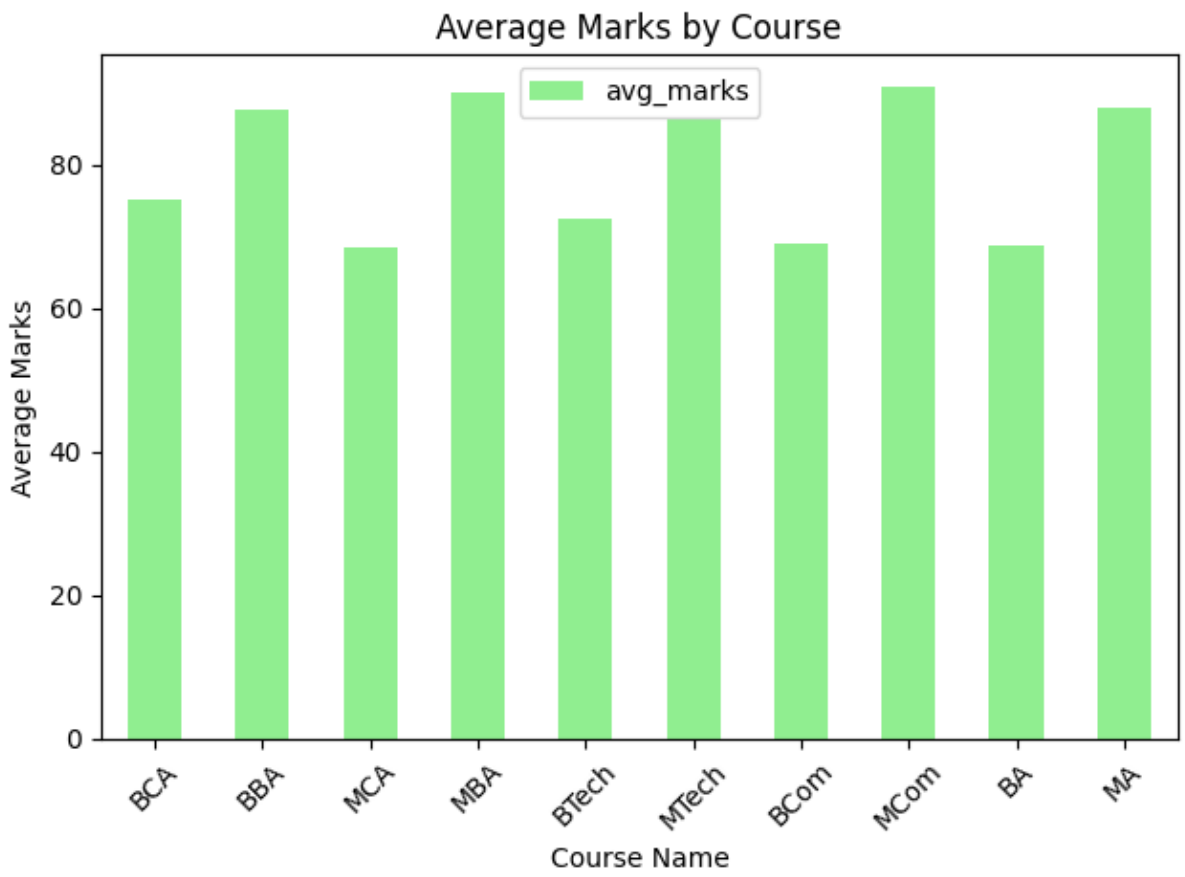
```
C:\Users\sachi\AppData\Local\Temp\ipykernel_19848\363640
  df_students = pd.read_sql(query, conn)
```

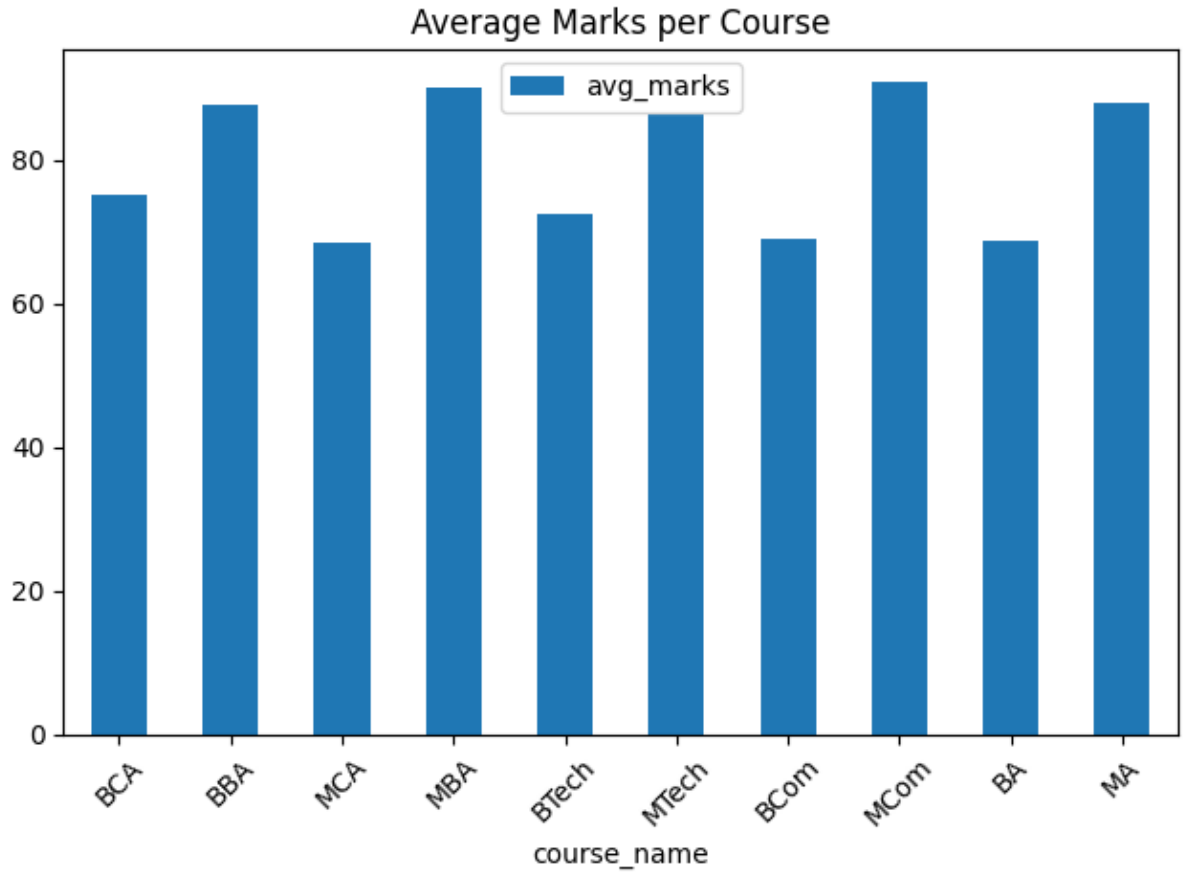
	student_id	name	age	course	city
0	1	Nidhi	21	BCA	Dehradun
1	2	Rahul	22	BCA	Delhi
2	3	Aman	20	BCA	Mumbai

Graph


Top 5 Students by Marks
marks
Marks
0
20
40
60
80
Rashmi
Anjali
Komal
Kriti
Shreya
Student Name


Students with Attendance Below 75%
attendance_percentage
Attendance Percentage
0
10
20
30
40
50
60
70
Rahul
Karan
Rohit
Vikas
Arjun
Sahil
Deepak
Varun
Ajay
Nitin
Rakesh
Mohit
Tarun
Gaurav
Harsh
Kunal
Abhishek
Suresh
Ankit
Raj
Vivek
Student Name


Average Marks by Course
avg_marks
Average Marks
0
20
40
60
80
BCA
BBA
MCA
MBA
BTech
MTech
BCom
MCom
BA
MA
Course Name


Average Marks per Course
avg_marks
0
20
40
60
80
BCA
BBA
MCA
MBA
BTech
MTech
BCom
MCom
BA
MA
course_name